Module5Lab2

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1 Module 5 Lab 2: Textbook Problem 5.3

1.0.1 Housing Type Bar Chart

Name: Mac Mckay

Course: CS 82A

Repo: https://github.com/raulmckay-blip/cs82a-portfolio

1.1 Overview

This project takes the raw customer data, removes the rows with a missing housing type, and charts how many customers fall into each housing type.

1.2 Load the data

The file is tab-separated, so I added `sep='\t'`. Without it pandas looks for commas and puts everything in one column.

[2]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_theme(style='whitegrid')

cust = pd.read_csv('custdata.tsv', sep='\t')
print(cust.shape)   # 1,000 customers, 11 columns
cust.head()
```

```
(1000, 11)
```

[2]:

	custid	sex	is.employed	income	marital.stat	health.ins	housing.type	recent.move	num.vehicles	age	state.of.res
0	2068	F	NaN	11300	Married	True	Homeowner free and clear	False	2.0	49.0	Michigan
1	2073	F	NaN	0	Married	True					
2	2848	M	True	4500	Never Married	False					
3	5641	M	True	20000	Never Married	False					
4	6369	F	True	12000	Never Married	True					

```
1                    Rented        True           3.0  40.0      Florida
2                    Rented        True           3.0  22.0      Georgia
3     Occupied with no rent       False           0.0  22.0   New Mexico
4                    Rented        True           1.0  31.0      Florida
```

1.3 Zoomin in: The housing types

[3]:
```
cust['housing.type'].value_counts(dropna=False)
```

[3]:
```
housing.type
Homeowner with mortgage/loan    412
Rented                          364
Homeowner free and clear        157
NaN                              56
Occupied with no rent            11
Name: count, dtype: int64
```

Before removing anything, I counted every housing type including the missing ones. There are four real types plus 56 missing. The file writes them as `NA`, and pandas reads that as missing on its own, so I didn't have to convert anything.

1.4 Remove the NA rows (and count them)

The book's hint says to subset with a condition on the housing-type field, so I filtered for rows where `housing.type` is not missing (`.notna()`). I filtered on just that column on purpose. When I checked, running `dropna()` on the whole table would also drop rows missing other columns (like `is.employed`, with 328 missing) and leave only 664 customers.

[4]:
```
print('Missing per column:')
print(cust.isna().sum())
print()
print('Rows left if I dropped NA in ANY column:', len(cust.dropna()))
```

```
Missing per column:
custid            0
sex               0
is.employed     328
income            0
marital.stat      0
health.ins        0
housing.type     56
recent.move      56
num.vehicles     56
age               0
state.of.res      0
dtype: int64

Rows left if I dropped NA in ANY column: 664
```

[5]:
```
before = len(cust)
housing = cust[cust['housing.type'].notna()]   # keep only rows that have a␣
 ↪housing type
after = len(housing)
dropped = before - after

print(f'Before: {before} rows')
print(f'After:  {after} rows')
print(f'Dropped {dropped} rows ({dropped / before:.1%}) with no housing type')
```

```
Before: 1000 rows
After:  944 rows
Dropped 56 rows (5.6%) with no housing type
```

[6]:
```
# make sure nothing missing is left
print('NA left:', housing['housing.type'].isna().sum())

counts = housing['housing.type'].value_counts()
counts
```

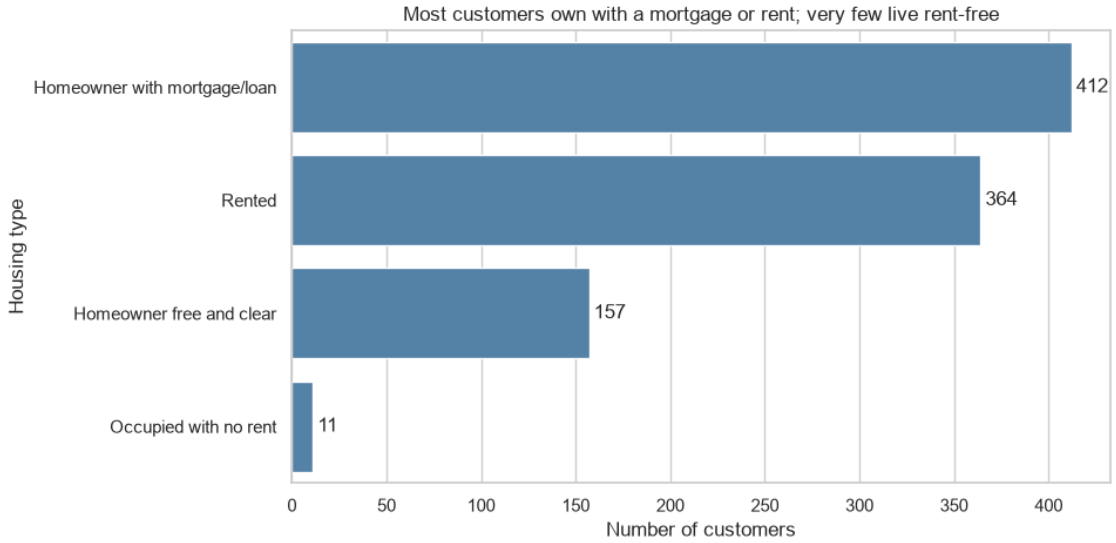
```
NA left: 0
```

[6]:
```
housing.type
Homeowner with mortgage/loan    412
Rented                          364
Homeowner free and clear        157
Occupied with no rent            11
Name: count, dtype: int64
```

1.5 Sorted bar chart

I made the bars horizontal because the housing type names are long and overlapped on the x-axis.

[7]:
```
plt.figure(figsize=(9, 5))
ax = sns.countplot(data=housing, y='housing.type', order=counts.index,␣
 ↪color='steelblue')
ax.bar_label(ax.containers[0], padding=3)   # show the count on each bar
plt.title('Most customers own with a mortgage or rent; very few live rent-free')
plt.xlabel('Number of customers')
plt.ylabel('Housing type')
plt.show()
```



Homeowners with a mortgage or loan are the biggest group (412), with renters close behind (364), then people who own their home outright (157). Only 11 customers live somewhere without paying rent.

1.6 Rows excluded

I excluded 56 of the 1,000 rows (5.6%) because their housing type was NA, so the chart is based on 944 customers.

1.7 AI Use

I used AI to help with syntax and refine wording on this lab.